Lopcombe Corner — Does the Level at the Start of Winter Matter?

Winter peak vs winter rainfall with and without the start-of-winter groundwater level · 25 logger-era winters 2000/01–2024/25 · levels: EA logger E14150 (corrected winter peaks) · rainfall: HadUK-Grid 1 km, borehole cell · analysis 20 Oct 2026

Bottom line. Yes, but modestly. Adding the start-of-winter level to Oct–Mar rainfall lifts the share of winter-peak variation explained from 0.80 to **0.85** and cuts the typical error from 3.3 m to 2.9 m. Each extra metre of starting level adds about **3.6 m** to the peak (p = 0.02). The effect is small in practice because the well drains back to almost the same level every autumn (78.49 ± 0.44 m), so rainfall remains by far the main driver.

1. What the start level is

- The start level is the autumn recession low of the logger record before winter recharge begins. The recession is flat, so it is essentially the level on about 1 October. Values were extracted on 7 Sep 2026.
- Over the 25 winters it averaged 78.49 m (standard deviation 0.44 m), ranging from 77.58 m (2006/07) to 79.36 m (2008/09).
- It is unrelated to that winter's rainfall (r = 0.07), so it adds independent information. On its own it correlates with the peak at r = 0.27.

2. How the finding has developed

Analysis (date)	Rainfall used	Without start	With start	Start effect
v1 (7 Sep)	Broughton Oct–Jan, first-pass peaks	0.52	0.56	partial r 0.28
v2 (7 Sep)	Broughton Oct–Dec, first-pass peaks	0.43	0.49	not significant
Correction (7 Sep)	Broughton Oct–Jan, corrected peaks	0.55	0.57	—
v3 (7 Sep)	Broughton Oct–Jan, corrected peaks	0.64	0.73	—
v3 (7 Sep)	Broughton Oct–Mar, corrected peaks	0.74	0.83	≈ +5 m per m
This report	HadUK-Grid Oct–Jan, corrected peaks	0.76	0.81	+3.7 m per m, p 0.03
This report	**HadUK-Grid Oct–Mar, corrected peaks**	**0.80**	**0.85**	**+3.6 m per m, p 0.02**

Columns 3–4 are R^2 (share of variation in winter peak explained). The September Oct–Jan numbers vary because of the peak correction and the re-summed Broughton totals.

- Early analyses found little or no start-level effect because they used a rainfall window that missed Feb–Mar recharge, first-pass peaks with three truncated winters, and Broughton totals that under-read wet winters.
- Once those were fixed (v3), the effect became clear. With the more accurate HadUK-Grid rainfall, rain alone explains more (0.80 vs 0.74), so there is less left for the start level to add (+4 points vs +9), but the effect remains statistically significant.

3. Current model (HadUK-Grid Oct–Mar rainfall)

Model	Term	Coefficient	Std error	p-value
Rain only	Intercept	65.378	2.500	< 0.001
	Oct–Mar rain (per mm)	0.043	0.004	< 0.001
Rain + start level	Intercept	-220.049	113.365	0.065
	Oct–Mar rain (per mm)	0.042	0.004	< 0.001
	Start level (per m)	3.641	1.446	0.020

Fit: peak ≈ -220.0 + 0.042 × Oct–Mar rain + 3.64 × start level. F-test for adding the start level: F = 6.34, p = 0.020. Partial correlation of start level with the peak after allowing for rain: r = 0.47.

In practical terms: a start one standard deviation (0.44 m) above average raises the expected peak by about 1.6 m. By comparison, 100 mm more Oct–Mar rain raises it by about 4.2 m, and one standard deviation of Oct–Mar rain (127 mm over 1963–2025) by about 5.4 m.

4. Winter by winter

Residual = actual peak minus predicted. A residual closer to zero with the start level means it helped.

Winter	Oct–Mar mm	Start m	Peak m	Pred: rain	Pred: + start	Resid: rain	Resid: + start
2000/01	**819**	**78.65**	**101.39**	**100.6**	**101.0**	**+0.8**	**+0.4**
2001/02	488	78.55	85.33	86.4	86.7	-1.1	-1.3
2002/03	**671**	**78.71**	**100.79**	**94.3**	**95.0**	**+6.5**	**+5.8**
2003/04	475	78.30	82.72	85.8	85.2	-3.1	-2.5
2004/05	382	78.10	80.05	81.8	80.5	-1.8	-0.5
2005/06	390	77.70	79.33	82.2	79.4	-2.8	-0.1
2006/07	630	77.58	90.76	92.5	89.1	-1.7	+1.7
2007/08	449	78.92	85.76	84.7	86.3	+1.1	-0.6
2008/09	411	79.36	92.62	83.0	86.3	+9.6	+6.3
2009/10	604	78.17	88.15	91.4	90.2	-3.2	-2.0
2010/11	351	78.25	81.05	80.5	79.8	+0.6	+1.3
2011/12	268	77.99	78.40	76.9	75.3	+1.5	+3.1
2012/13	619	79.02	94.42	92.0	93.9	+2.4	+0.5
2013/14	**860**	**78.58**	**102.01**	**102.4**	**102.5**	**-0.4**	**-0.5**
2014/15	489	78.86	83.13	86.4	87.8	-3.3	-4.7
2015/16	524	78.56	88.15	87.9	88.2	+0.2	-0.1
2016/17	369	78.70	78.86	81.3	82.2	-2.4	-3.3
2017/18	464	78.47	89.54	85.3	85.3	+4.2	+4.2
2018/19	451	78.40	80.66	84.8	84.5	-4.1	-3.9
2019/20	**716**	**77.93**	**97.25**	**96.2**	**94.0**	**+1.1**	**+3.2**
2020/21	576	78.66	91.72	90.2	90.8	+1.6	+0.9
2021/22	414	78.67	79.61	83.2	83.9	-3.6	-4.3
2022/23	709	78.07	92.62	95.9	94.2	-3.3	-1.6
2023/24	**796**	**78.92**	**97.54**	**99.6**	**101.0**	**-2.1**	**-3.5**
2024/25	501	79.08	90.31	86.9	89.1	+3.4	+1.2

Bold rows reached the 97 m Pitton threshold.

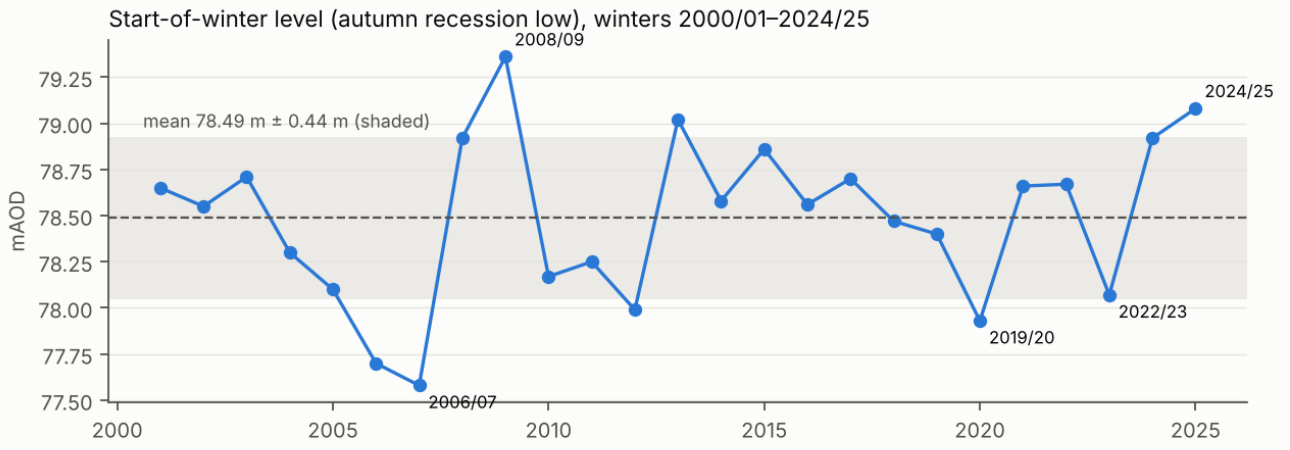
Where the start level helps — and where it doesn't

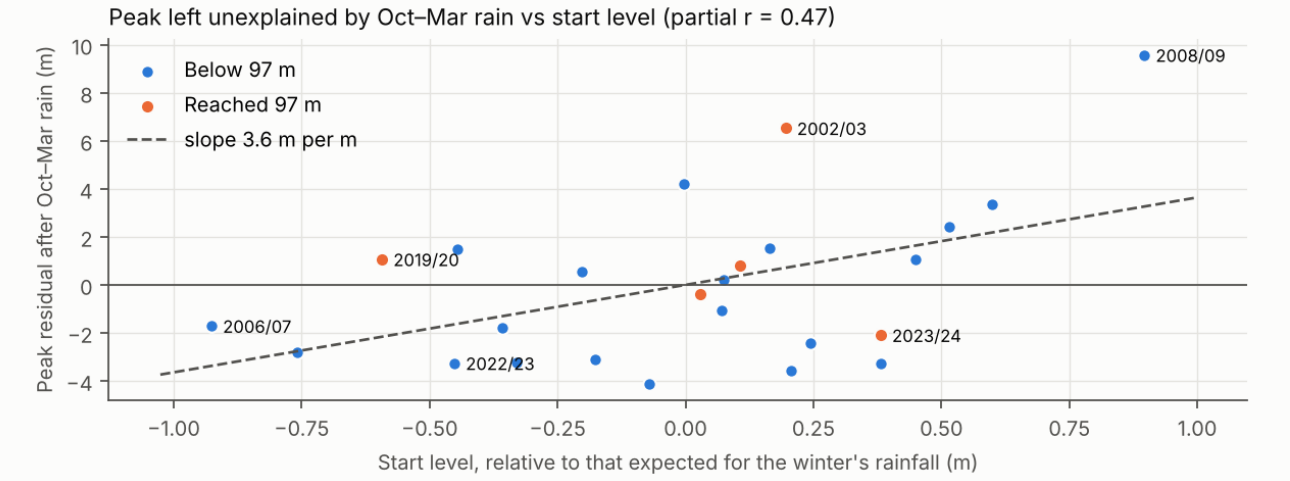
- **Helps most:** 2005/06 (–2.8 → –0.1 m) and 2004/05 (–1.8 → –0.5 m), both low-start winters; 2008/09 (+9.6 → +6.3 m), the highest start; 2012/13 (+2.4 → +0.5 m); 2022/23 (–3.3 → –1.6 m), a low start; 2024/25 (+3.4 → +1.2 m).
- **Makes it worse:** 2019/20 (+1.1 → +3.2 m) — it started low yet still reached 97.25 m, driven by the very wet February 2020 (Storms Ciara and Dennis); 2023/24 (–2.1 → –3.5 m) — a high start, but the peak came early in January; 2006/07 overshoots (–1.7 → +1.7 m); 2014/15 (–3.3 → –4.7 m).
- **Still unexplained by either model:** 2002/03 (+5.8 m), 2008/09 (+6.3 m) and 2017/18 (+4.2 m, an April peak from late rain). These point to rainfall timing within the winter as the next missing factor.
- **Flood-threshold calls are unchanged.** Both models flag the same winters above 97 m (2000/01, 2013/14, 2023/24) and both miss 2002/03 and 2019/20, whose peaks came from intense rain concentrated in a few weeks.

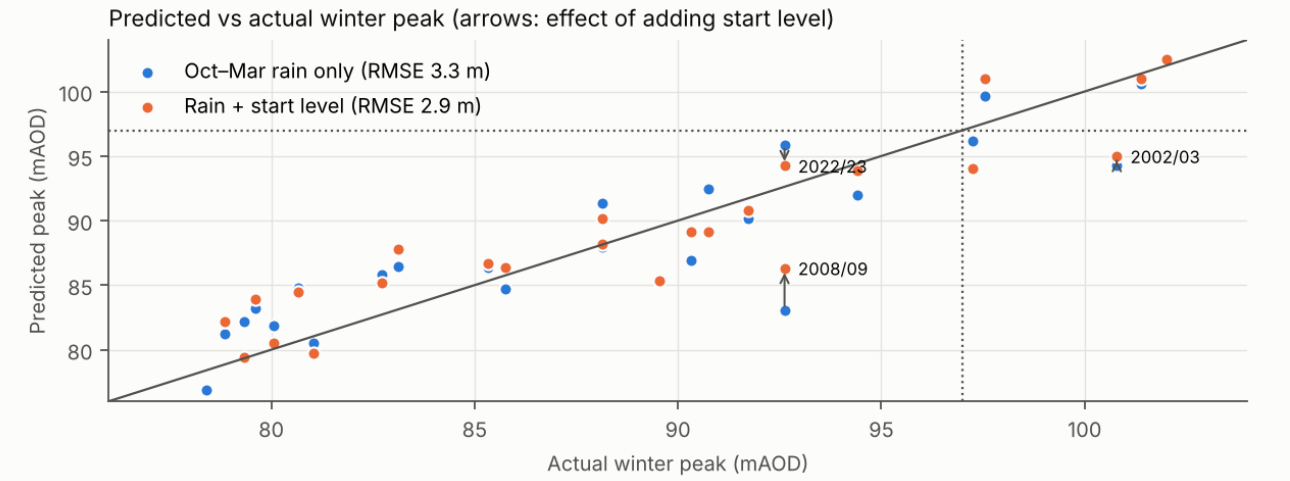
5. Caveats and next steps

- Only 25 winters: the start-level coefficient is uncertain (standard error 1.4 m per m; 95% range roughly 0.6–6.6 m per m). The direction is clear; the size is not.
- The start levels come from the first-pass logger extraction (7 Sep). They are recession lows, which the truncation problem did not affect, but they have not been re-extracted.
- Not yet done: extending to all 61 winters back to 1964 using autumn manual dips (needs EA API access), and adding a rainfall-timing term such as February rain or the wettest 30 days, which the monthly HadUK-Grid data can support now.

Reproduce: `python scripts/start_level_analysis.py` (reads output/haduk/lopcombe_winter_rain_vs_peak.csv; writes start_level_winters.csv, start_level_results.json and Lopcombe_start_level.png to output/haduk/).


Start-of-winter level (autumn recession low), winters 2000/01–2024/25
mAOD
mean 78.49 m ± 0.44 m (shaded)
2008/09
2024/25
2022/23
2019/20
2006/07


Peak left unexplained by Oct–Mar rain vs start level (partial r = 0.47)
Peak residual after Oct–Mar rain (m)
Below 97 m
Reached 97 m
slope 3.6 m per m
2008/09
2002/03
2019/20
2006/07
2022/23
2023/24
Start level, relative to that expected for the winter's rainfall (m)


Predicted vs actual winter peak (arrows: effect of adding start level)
Predicted peak (mAOD)
Oct–Mar rain only (RMSE 3.3 m)
Rain + start level (RMSE 2.9 m)
2022/23
2002/03
2008/09
Actual winter peak (mAOD)